NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

L26.12 Sparse Autoencoder Interpretability Transfer

Group Rapid Trials Lab

Status Closed (negative result)

Period 2026-10-24 23:22 — 2026-11-11 21:29

Stage conclusion

Project report

L26.12 Transfer of sparse autoencoder interpretability

Approval: the PI approved A289 on 10-26, pplan approve. Try R23 Pangu first. Write scope , .

Check of the original paper (step 0, 10-26)

Original paper: Liu, Li, Sun, Ni 2026, arXiv 2610.28904v1 (submitted to GRL), 22 pages. Zotero GV6PQBKB / attachment DVIHW7UE; project copy. All page numbers below are PDF pages (matching the manuscript footer numbering).

Item checked	What the paper does	Source
Model and layer	GraphCast 0.25° (trained on ERA5 1979–2017); 16 processor layers, each with 40,962 grid nodes × 512-dimensional hidden states; main analysis layer is Layer 8	PDF p.5
SAE configuration	TopK SAE, dictionary 4,096, K=32; architecture follows MacMillan & Ouellette 2025; 5 seeds per layer (50108–50112), 80 dictionaries in total	PDF p.5, p.9
Time split for SAE training/validation/analysis	SAE trained on 1979–2017 hidden states (no tropical cyclone labels, unsupervised), reconstruction validated on 2018–2021; tropical cyclone analysis uses only 2018–2019	PDF p.5
Storm split	**No storm-level training/test split.** All 205 storms, 2,115 time steps and 4,118 storm–time-step records are used in a single 2018–2019 analysis; feature **selection** (Top-1/Top-4 by median F1) and **evaluation** of the attribute correspondence are done on the same storms and the same time steps	PDF p.5, p.8–9
Post hoc selection	The Top-4 count was "chosen exploratorily after viewing the full picture of the reference dictionary," then fixed for all 80 dictionaries	PDF p.9–10
Attribute definitions	Position: F1 between the feature's positive activation region and the IBTrACS center + R34 mask; intensity: difference D between the minimum ERA5 sea-level pressure anomaly within the mask (not the IBTrACS maximum wind) and the maximum activation within the feature mask, each min–max normalized; structure: spatial Spearman ρ between feature activation and −p′ within ±12° (within-storm spatial correlation, not cross-storm intensity correlation)	PDF p.5–8
Main numbers	Layer-8 reference dictionary f613: median F1 0.512, D 0.180, ρ 0.644; Top-1 median F1 across 5 seeds 0.345–0.512	PDF p.9, p.11
Any intervention/ablation	**None.** The authors state that reading out only shows "correspondence," not "use"; whether the model relies on it requires intervention or ablation (pointing to MacMillan & Ouellette 2025)	PDF p.13

Item checked	What the paper does	Source
Limitations acknowledged by the authors	Sensitivity to SAE dictionary size, sparsity and Top-k subset size not tested; only one model, two years, single step	PDF p.13
Public materials	SAE checkpoints, analysis code, processed results: Zenodo 10.5281/zenodo.23492703; ERA5 inputs taken from the public WeatherBench 2 zarr (GCS)	PDF p.14–15

Reading: the paper's conclusion goes only as far as "correlation/correspondence," and selection and evaluation use the same set (selection bias not quantified); correlation between time steps is not handled (about 10 time steps per storm on average). These affect only our assessment of the paper, not this project's criteria: this project splits by storm + year and performs interventions (launch plan §2). **Two points useful for this project**: ① the WeatherBench 2 ERA5 zarr is publicly readable, which provides a ready source for "ERA5 cloud copy to be checked" in launch plan §5/§6; ② the precedent for intervention is MacMillan & Ouellette 2025 (arXiv 2512.24440, retrieved, Zotero ANNAEAZP); the step 3 intervention method follows it.

Literature retrieval (10-26)

Paper		Result
Liu et al. 2026, arXiv 2610.28904	Confirmed missing	Retrieved, Zotero GV6PQBKB
Cunningham/Huben et al. 2023/2024, arXiv 2309.08600	Confirmed missing	Retrieved, Zotero TAZDXFZX
Gao et al. 2024, arXiv 2406.04093 (TopK SAE)	Confirmed missing	Retrieved, Zotero Q7T83PB8
MacMillan & Ouellette 2025, arXiv 2512.24440 (GraphCast SAE + intervention)	Confirmed missing	Retrieved, Zotero ANNAEAZP (added beyond the launch plan: intervention precedent)
Bricken et al. 2023 Towards Monosemanticity	Confirmed missing	Published only as a web page on transformer-circuits.pub, no DOI, no arXiv, not supported; passed to litwiki to decide whether a card can be built from the web page
Templeton et al. 2024 Scaling Monosemanticity	Confirmed missing	Same as above

Tried and results

Step	Time	Result	Evidence
Step 0 check of the original paper	10-26	No storm split (selection/evaluation on the same set); no intervention	"Check of the original paper" on this page
Step 1 truncation gate	10-26 16:35–16:51	**Passed**: pangu_weather_24.onnx (ECMWF mirror, sha256 …) was cut into two segments at node 3856/7717, with the boundary being the backbone (1×131040×384, coarse-resolution layer) + encoder skip connection (1×521280×192); the maximum relative difference between the two segments in series and the full network was 0 for both outputs (criterion <1e-4). Full-network sanity: 24 h forecast initialized 2019-09-01 12Z, MSL RMSE 78 Pa (persistence 598), Z500 RMSE 137 m^2/s^2 (persistence 564). Extra: scaling the whole skip-connection tensor by 1.01 gave finite output, maximum MSL change 33 Pa	; first run had a graph-construction error (constants as float inputs), see

| Steps 2–3 correlation + intervention | 10-26 16:57–19:16 | Training 275 storms / test 46 storms in 2021 (1 time step per storm, sample reduced post hoc; WB2 actually goes only to 2021); SAE width 1536, K=64; ① main reading not met (best lower bound 0.660 < permutation maximum 0.679), parallel readings ① and ② met but ③ not met; divergence 0/3900; pure decoder replacement α=1 median 0.995 hPa |; draft conclusion v1 |

| Options B, C (L26.12-02) | 10-26 20:31–21:08 | B: same program reproduced the original results; 3 seeds, permutation 1,000 / bootstrap 5,000, main reading ① not met in any (991 difference −0.015 to −0.054), range of permutation extremes across seeds 0.033–0.082; C: added 12 storms (4 without a t+24 position), 42 storms in total, ③ judgment unchanged, target median |E| ≤0.064 hPa, divergence 0/5,460 | |

| Option A (L26.12-03) | 11-03 10:44–12:22 | Cut point [local path omitted] (7651/7717), gate passed; SAE width 1536, K=64; pressure/286 test |ρ| 0.831 passed main reading ① but not ② (PCA 0.886); wind/286 and other parallel readings passed ① and ②; ③ failed in all cases (minimum-pressure change under intervention all 0, structural); (i) 20-seed difference [−0.051,+0.042] spans 0 | |

Resource ledger

Time	Item	CU
10-26	Step 0 (no compute)	0
10-26 16:35–16:51	Step 1 gate, A100 session l2612-gate about 16.3 min (including one failed first run), stopped	≈1.5 (converted at 5.4 CU/h)
10-26 16:57–19:16	Steps 2–3, session l2612-sae-step2 about 2.31 h, stopped	≈12.5 (converted at 5.4 CU/h)
10-26 20:31–21:08	Options B, C, session l2612-bc-step3, ≤0.62 h, stopped	≤3.34 (keepalive did not perform the final pull: keepalive recorded stopped, final_pulled empty; results are based on the local step3 files: interventions_remaining.jsonl 1572 lines, last progress line done/12 storms, b_results.json readable; checked by the lead on 10-26)
11-03 10:44–12:22	Option A, session l2612-postmerge-stepA 1.64 h, stopped	≈8.9
11-10 00:05–(reclaim time unknown, ≤01:29)	Option H, session l2612-stepH, reclaimed after the 00:19 platform shutdown	≈1.4–7.6
11-10 23:29–11-11 00:33	Option H′, session l2612-stepH2 1.07 h, stopped (40/42 storms)	≈5.8
	Cumulative	**≈33.4–39.6 / stop line 39 / cap 40**

Parameter source table

Parameter	Value
Pangu weights	pangu_weather_24.onnx, 1,181,711,187 B, sha256 613a5c140a1399abcaffb4dbce32af373a1f5f56c515704f5be61925bb9fdcfd
ERA5 input	WeatherBench 2 0.25° 6 h zarr, latitude flipped to 90→−90
SAE dictionary multiple, K	Launch plan states 4–8×, TopK; original paper 4096/512 = 8×, K=32

Literature basis (cards built by litwiki on 10-26)

- Original paper : GraphCast, features selected and evaluated on the same 2018–2019 storms, no holdout, no intervention (consistent with "Check of the original paper" on this page).
- Intervention precedent (PDF pp. 7–9): GraphCast layer 8, only the f3243 coefficient is multiplied by 1+γ and the reconstruction residual is added back; this is the same approach as this project's L26.12-01 h' = h + (α−1)·z_k·d_k (α=1+γ). Its γ=+0.5 produced runaway, nonphysical responses in a 10-day, 40-step autoregressive forecast with re-application at every step. This project is a single 24 h step applied only once, but α=2 (γ=+1) exceeds its largest amplitude; divergence counts toward the "not judged" condition under the criteria and must be reported for each α.
- Methods: , (TopK definition and choice of K and width); web versions , .

Stop category

11-11: negative result (this configuration only; intervention 40/42 storms, partial): six-reviewer ruling (A argued for keeping "blocked"; reasons for not adopting this are in the ruling). Earlier records follow: 11-03: after option A, still **blocked, awaiting the PI** (negative condition (i) not met; intervention at the post-merge layer has no resolving power); blocked report and lead recommendation in . The 10-26 record follows: **Blocked, awaiting the PI** (10-26 21:45, six-reviewer ruling). Under the locked criteria the verdict word is "abandon (this configuration only)," but two conditions are not met, resolving power (① within sampling noise, ③ no bound on the effect) and failure cause (skip connection not intervened on), so it is not registered as a negative result. Blocked report, option costs and lead recommendation in §5 and "Lead recommendation."

11-03 A304 approval

The PI: "approve all as recommended" → option A: retest at the layer after the skip-connection merge, cap 33 CU, stop at 30 CU, 0 SU, no extension to D or E. Decision ; dispatched to the sub-project.

Option A (11-03)

The PI approved on 11-03 per the A304 lead recommendation; task brief (operational definitions of negative conditions (i)(ii) added on 11-03 before the run); from 11-03 10:44, session l2612-postmerge-stepA: gate passed (cut point , 1×521280×384, segment B has only this one non-constant input, series error 0); from 11:10 the lead took over SAE/correlation/intervention (see the 11:10 section of the same decision file); intervention cut off at 12:45, follow-up check stopped the session at 12:49. Results written to . - 11-03 12:22: option A completed (session l2612-postmerge-stepA about 8.9 CU, cumulative about 26.2 CU, stopped): at the post-merge layer, ① pressure main reading passed and wind parallel reading passed; ② wind passed, pressure failed; ③ all 0 for 42 storms, structurally no resolving power; negative condition (i) interval [−0.051,+0.042] spans 0. Still blocked

(awaiting the PI); not sent for six-reviewer review. Lead recommends H (about 4–6 CU, requires stop line 30→32). Conclusion v4 .

Option H (11-10)

The PI on 11-10: "go ahead as you recommended." **At 00:19 the workstation user services shut down entirely, keepalive was stopped, the VM was reclaimed, and H was interrupted** (decision , conclusion v5). Decision ; scripts and results ; intervention self-stopped at 01:05, follow-up check stopped the session at 01:09.

Option H′ rerun (11-10 23:29)

The PI on 11-10: "OK, go as recommended; from now on, CRC computations this small need no approval." Decision ; results written to ; intervention self-stopped at 00:26, fallback stop of the session at 00:32.

Wrap-up (11-11)

- Run-root cleanup: all computation for this project was on Colab, and all sessions have been stopped/reclaimed; **there is no CRC/TACC run root**, so no cleanup is needed. The L26.4 data on CRC were deleted long ago and are unrelated to this project. Local about 67 MB in total (window archives, features, result tables), kept.
- Conditions for retrying are in conclusion v7 §5.

Key figures